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PI Name:	Grigoriev, Roman Ph.D.		
Project Title:	Optical Manipulation of Dynamic Contact Lines: Experiments and Theory		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schatz, Michael (Georgia Tech)		
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Task Description:	We propose studies of driven thin liquid films that exhibit wetting/spreading and capillary-driven phenomena cited by the National Research Council (NRC) as 'high-priority' research directions for NASA's Office of Biological and Physical Research (OBPR). The proposed study will address unsolved problems involving moving contact lines by combining theoretical modeling with experiments that can probe flow dynamics in ways that heretofore were not possible. This study uses a newly-developed experimental technique to manipulate optically (via the Marangoni effect) a broad class of capillary-dominated flows simultaneously at multiple spatial locations. With this technique, well-defined initial conditions can be imposed on spreading liquid films to probe dynamics and feedback can be applied to enable control. The experiments will examine spreading driven by Marangoni effects, which are ubiquitous in microgravity. Moreover, experiments will also examine gravitationally-driven spreading on inclined surfaces to gain insight into effects that may arise in Human Exploration and Development of Space (HEDS) technologies designed to operate in reduced gravity on other planets or in 'artificial' gravity produced by rotation. For all cases, optical actuation will be used to perturb the spreading films using disturbances with well-defined wave numbers, thereby permitting quantitative measurement of the growth/decay rates. Our ability to impose well-controlled disturbances in experiments will also be used to measure transient amplification effects, thereby enabling, for the first time in any fluid-dynamic system, quantitative theoretical/experimental examination of the controversial role that transients play in instability and transition. These results will be compared with predictions from different lubrication models using both linear stability analysis via eigenvalues and generalized stability analysis to characterize predicted transient growth due to the 'non-normal' character of linear operators that describe spreading films, allowing resolution of currently existing discrepancies between experimental measurements and theoretical predictions of growth rates and transient amplification factors. Additionally, both theory and experiment will be used to investigate the role of nonlinear terms in transition and the relation between transient amplification and secondary instability of a weakly perturbed flow, which offer alternative explanations for transition. Finally, optical actuation will be used to provide feedback in response to flow measurements, thereby permitting investigation of methods for selecting desired flow states, either by selective patterning of the flow or by suppression of instability. The results of the proposed work should contribute strongly to the fundamental, strategic and commercial research thrusts of OBPR. Our work emphasizes the use of optical actuation as a unique tool for studying fundamental scientific problems in liquid films that not only address the OBPR Physical Sciences Research Division's Fluid Physics interests in dynamics, instabilities and interfacial phenomena, but also connect, in novel ways, to broader issues in Fluid Physics, including the transition to turbulence. The results from the proposed work will also be strategically important for evaluating the potential for success or failure of proposed HEDS technologies. The dynamics of wetting and spreading (recently ranked by the NRC as 'most important' in strategic impact relative to other high priority areas across all microgravity disciplines) play a crucial role in liquid flow management in a vast array of HEDS technologies for power generation, propulsion, and life support. The spreading behavior of liquids is of central importance in applications as diverse as the storage of cryogenic propellants, the suppression of fire aboard spacecraft, the rewetting of hot surfaces in space-based multiphase heat transfer devices and the cleanup of liquid spills in low gravity. The results from the proposed work should also be commercially relevant to devising practical methods of controlling thin film flows; in particular, our results could lay the foundation for a new generation of devices for controlling interfacial flows on earth as well as in space. Wetting and spreading dynamics (ranked by the NRC as 'most important' in commercial impact relative to other high priority areas across all microgravity disciplines) can strongly affect the coating of solid surfaces, a process that is ubiquitous in industry (e.g., spin-coating of semiconductor wafer) and are directly relevant to emerging technologies such as microfluidics devices for biological and chemical applications. Thus, in summary, our proposed work closely follows the recommendations of expert external committees by building upon long standing OBPR/PSRD strengths in directions that contribute most effectively and directly to OBPR/PSRD's support of NASA's mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Liquid spreading/wetting phenomena play an important role in numerous commercial terrestrial applications. For instance, contact-line dynamics controls the coating of solid surfaces, a process that is ubiquitous in industry (e.g., spin-coating of semiconductor wafers). Improved understanding of stability of driven contact lines will help impove the speed and quaility of the coating process. Another important application is to “emerging technologies” such as microfluidics, where termocapillary and electrowetting effects have been proposed as a basic physical mechanism for manipulating minute amounts of fluids. Our research paves the way to a new generation of dynamically reprogrammable microfluidic devices where the flow is controlled using optical means. Finally, through the development of mathematical apparatus, our research paves the way to improving fundamental understanding of stability of high-Re shear flows and "bypass" transitions to turbulence.
Task Progress:	The physics of triple-phase contact lines, where solid, liquid, and gaseous phases meet, is a very old subject. Although the static isothermal (e.g., thermodynamic equilibrium) problem is well understood, the corresponding out-of-equilibrium problem characterized by non-vanishing mass or heat fluxes still holds many surprises. Our combined theoretical and experimental studies have concentrated on the stability of dynamic contact lines in gravitationally driven (isothermal) and thermocapillary driven (non-isothermal) flows of non-volatile thin films. The major goal of our research was to extend the predictive power of existing models of thin film flow beyond linear stability analysis which is only accurate for very small disturbances and long time scales. Qualitatively different dynamics can be observed in experiments, for moderate disturbance amplitude and short time scales. We constructed a proper theoretical description of short-term dynamics of disturbances relative to the state describing transversely uniform spreading of fluid and showed that generic disturbances initially experience transient growth, even when the uniform state is linearly stable. We also demonstrated that the disagreement in theoretical predictions for the magnitude of transient amplification of disturbances obtained using different models of gravity-driven films was due to their improper analysis in many studies. We established that the contact line dynamics is most sensitive to disturbances, such as surface roughness or chemical heterogeneity, that lie ahead of the contact line. We also found that the transient amplification magnitude depends sensitively on the streamwise and transverse structure of disturbances and that some disturbances could be amplified by many orders of magnitude. Several predictions of the theory have been confirmed through experiments on gravity and thermocapillary driven films. Large transient growth raises an interesting possibility that even for linearly stable states and very small disturbances an initially uniform flow can undergo an instability and develop a pattern of transverse modulation of the contact line position and film thickness, a phenomenon similar to “bypass” transition to turbulence in high-Re shear flows. We have investigated the role of nonlinear terms in the evolution of finite-amplitude disturbances and discovered that, for gravity driven flows nonlinearity can enhance the instability leading to pattern formation in the linearly stable regime.

	Beyond the contact line dynamics, our analytical weakly nonlinear model describing the dynamics of finite-amplitude disturbances predicted many features found in direct numerical simulations of high-Re shear flows. One such feature is the emergence, in the immediate vicinity of the stable uniform state, of unstable equilibrium solutions analogous to “lower branch” coherent states such as the Tollmien-Schlichting waves that undergo secondary instability in boundary shear flows. While their importance in the context of high-Re shear flows is only speculated, we have shown that these unstable states do indeed play a key role in the pattern formation mechanism anchoring the boundary between the sets of initial conditions leading to patterned/unpatterned final states. To summarize, our studies have improved fundamental understanding of contact line dynamics. They also developed a mathematical apparatus that can be used to study transitions between laminar and turbulent flows at high Re numbers.
Bibliography Type:	Description: (Last Updated: 01/03/2008)
Abstracts for Journals and Proceedings	Grigoriev RO. "Non-normal evolution operators and transient amplification in thin liquid films." Conference on Differential and Difference Equations and Applications, Melbourne, FL, August 1-5, 2005. Conference proceedings distributed at the meeting, Conference on Differential and Difference Equations and Applications, August 2005. , Aug-2005
Abstracts for Journals and Proceedings	Schatz MF, Rieser J, Grigoriev RO. "Transient amplification of contact line instability: experiments and theory." Theoretical Aspects of Pattern Formation Workshop, University of Surrey, England, September 19-23, 2005. Conference proceeding distributed at the meeting, Theoretical Aspects of Pattern Formation Workshop, September 2005. , Sep-2005
Abstracts for Journals and Proceedings	Schatz MF, Rieser J, Grigoriev RO. "Transient amplification of contact line instability: experiments and theory." Workshop on Dynamical Systems and Turbulence, University of Warwick, England, October 31-November 2, 2005. Conference proceeding distributed at the meeting, Workshop on Dynamical Systems and Turbulence, October 2005. , Oct-2005
Abstracts for Journals and Proceedings	Rieser J, Grigoriev R, Schatz M. "Time evolution of transients in a thin liquid film." 58th Annual Meeting of the Division of Fluid Dynamics, American Physical Society, Chicago, IL, November 20-22, 2005. Proceedings distributed at the meeting, 58th Annual Meeting of the Division of Fluid Dynamics, American Physical Society, November 2005. , Nov-2005
Abstracts for Journals and Proceedings	Mitchell R, Grigoriev R. "Transient amplification and nonlinear stability of driven thin films." 60th Annual Meeting of the Division of Fluid Dynamics, American Physical Society, Salt Lake City, UT, November 18-20, 2007. Proceeding distributed at the meeting, 60th Annual Meeting of the Division of Fluid Dynamics, November 2007. , Nov-2007
Articles in Peer-reviewed Journals	Grigoriev RO. "Transient growth in driven contact lines." Physica D 2005 Sep 15;209(1-4):105-16. http://dx.doi.org/10.1016/j.physd.2005.06.015 , Sep-2005
Articles in Peer-reviewed Journals	Handel A, Grigoriev RO. "Transient dynamics and nonlinear stability of spatially extended systems." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Sep;74(3 Pt 2):036302. PMID: 17025738 , Sep-2006